

TRANSPORT SYSTEM IN HUMAN AND PLANTS

You have studied in previous chapter that the food we eat, gets digested, absorbed and distributed to all parts of the body. Similarly, you have also studied that we take in oxygen gas, in the breathing process, which also gets distributed to all body parts. Have you ever thought how the oxygen required by living things, move into their bodies and to the parts of the organisms where it is needed?

Do you ever wonder how waste product of the breathing process, 'carbon dioxide', is removed from our bodies?

In this Chapter you will learn about:

- Transport in Humans (Circulatory System, Heart and Blood Vessels)
- Common disorders (diabetes, heart problems and asthma)
- Transport in Plants
- Translocation in Plants

All the student will be able to:

- ✓ Explain the transport system in humans.
- ✓ Describe the structure and function of heart and blood vessels.
- ✓ Explain the working of circulatory system.
- ✓ Identify scientific developments that provide alternatives for dysfunctional body parts such as artificial tissues and organs; and their transplantation.
- ✓ Find out some disorders in human transport system can be affected by diet.
- ✓ Describe absorption of water in plants through roots.
- ✓ Explain how the structure of root, stem and leaves of a plant permit the movement of food, water and gases.



Figure 2.1 Human Transportation System

TRANSPORT EXPLORATION

Are diffusion and osmosis not sufficient enough to transport materials inside the multicellular organisms like humans? Why?

TRANSPORT IN HUMAN CIRCULATORY SYSTEM:

- ✓ Explain the transport system in humans.
- ✓ Describe the structure and function of heart and blood vessels.
- ✓ Explain the working of circulatory system.

Our body needs food and oxygen to produce energy and along with energy, waste materials are also produced in our body. How do all these materials move in our body? To help in movement of all these in the body, there is a need of transport system in human.

“The supply of food, water, oxygen and collection of waste materials such as carbon dioxide from our body is called transportation.”

The transport system in humans is called the blood circulatory system.

(1) Blood Vessels:

The blood vessels transport blood to all parts of the body.

In humans there are three kinds of blood vessels; arteries, veins and capillaries. The blood leaves the heart in arteries and returns through veins. The blood vessels differ from each other in the structure of their walls. The main function of vessels is to transport blood throughout the body.

Arteries:

Arteries take blood away from the heart. The walls of arteries have a small lumen (internal space), with a thick

DO YOU KNOW?

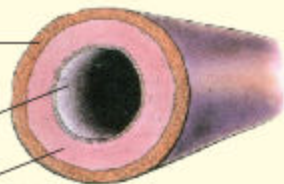
What is circulatory system?

This system involves the circulation of blood pumped out of a special organ called the heart through thousands of tubes called blood vessels and transporting fluid, blood.

BLOOD VESSELS

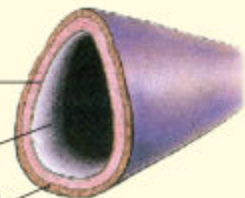
An artery

Thick outer wall
Small lumen
Thick layer of muscles



A Vein

Thin layer of muscles
Large lumen
Fairly thin outer wall



Do you find any difference in structural characteristics among these blood vessels?

A capillary

Very small lumen

Wall made of a single layer of cell



TRANSPORT EXPLORATION

Why is arterial diameter less than vein?

elastic and muscular structure, which can withstand the high pressure of the blood being pumped directly from the heart. The large arteries have greater proportion of elastic tissue which allows these vessels to hold up the high pressure caused by heart. While the smaller vessels called arteriole have less elastic tissue and more muscle fibres.

When these muscle fibres of the arteries contract, they make the passage in arteries smaller and restrict the flow of blood.

Arteries divide into very tiny blood vessels called capillaries.

Capillaries:

Capillaries are very tiny, microscopic and thin-walled vessels.

The thin walls of capillaries allow the exchange of oxygen, water, and nutrients from blood to cells. This also allows waste products from cells to pass into the capillaries. The walls of capillaries consist of single layer of cells which cause rapid diffusion of materials between cells and blood.

The capillaries gradually join up with one another to form veins.

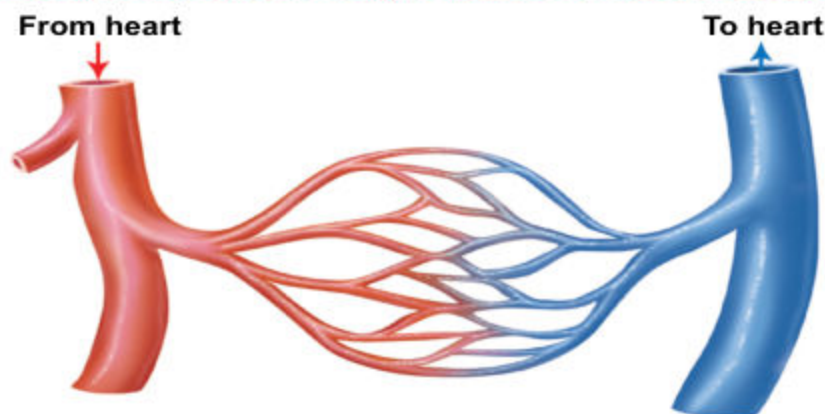
Veins:

All veins except pulmonary vein takes deoxygenated blood with less food and more carbon dioxide towards the heart. The walls of veins have larger lumen (internal space). They have thin wall, little elastic tissues and muscles that exert a lower pressure.

DO YOU KNOW?

All veins transport de-oxygenated blood to the heart except pulmonary vein carrying oxygenated blood.

Can you locate artery, capillaries and vein in figure 2.2?



INVESTIGATE

Can you guess the name of the structure which prevents the back flow of blood? This structure is normally present in veins.

Figure 2.2 Capillaries connect arteries to veins

(2) Heart:

Have you ever thought how the blood inside the blood vessels moves throughout the body? Which organ of the body forcefully pumps the blood in order to achieve maximum transportation? The heart is a hollow, cone-shaped and muscular organ; located between the lungs and behind the sternum (breastbone).

Two-thirds of the heart is located to the left of the midline of the body and one-third is to the right. The main purpose of the heart is to pump blood around the body.

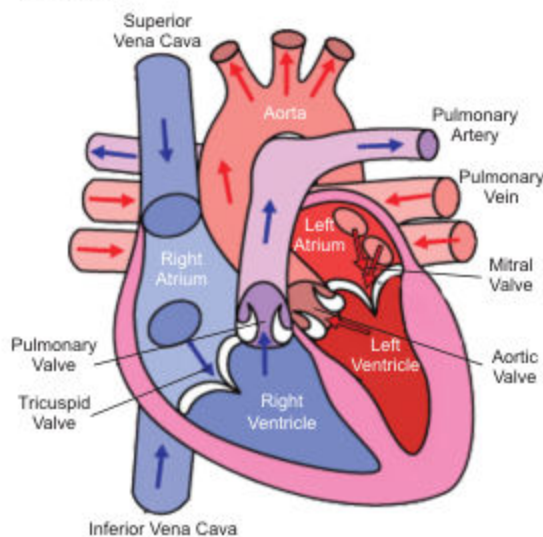


Figure 2.3 Human Heart

TRANSPORT EXPLORATION

Find out the name of arteries used to measure blood pressure and pulse?

DO YOU KNOW?

❖ What is blood?

Have you ever thought how the transportation of materials is possible within the system? This job can only be performed by transporting fluid, blood. Before discussing the heart, it is very important to explore the main features of blood:

- Our body contains four to five litres of blood.
- Blood is red in colour, both in arteries and veins but only in figures blood is shown with blue colour in veins to differentiate flow of oxygenated (rich in oxygen) and de-oxygenated blood (rich in carbon dioxide) in arteries and veins respectively.
- Blood is used to transport various substances like digested food, gases, chemicals and waste products from one part of the body to another by flowing continuously around the body.
- The heart is enclosed in a double membrane-sac called as pericardium.

❖ Blood Pressure:

Blood pressure is the pressure of blood exerted in the walls of the blood vessels. It is measured by sphygmomanometer. The normal value of human blood pressure is 120/80 mm Hg.



❖ Pulse:

Pulse is the rhythmic contraction and relaxation of arteries. Normally it is 72 times pulse per minute.

The heart is divided into four chambers separated by right and left sections by a septum (partition wall). The septum keeps blood separated from each of the side. The right side of the heart receives de-oxygenated blood from all parts of the body and pumps the blood towards the lungs whereas the left side of the heart receives oxygenated blood from the lungs and pumps the blood to all body parts.

DO YOU KNOW?

Do you know coronary artery emerges from the beginning of the aorta, near the top of the heart; supplies blood to the heart muscles for its proper functioning.

The two smaller upper chambers with thin walls receive blood through veins are called atria. The two lower chambers with thick walls receive blood through atria are called ventricles.

TRANSPORT EXPLORATION

Ventricles have thick muscular walled than atria. Why?

Right atrium receives de-oxygenated blood from the superior vena cava (collects blood from upper parts of the body) and inferior vena cava (collects blood from lower parts of the body). Left atrium receives oxygenated blood from lungs through pulmonary vein.

Have you ever thought what would happen when atria contracts?

Does blood flow back from the ventricles into the atria?

Blood never flows back towards atria because of valves. Four sets of valves are found in the human heart. Two valves are present between atria and ventricles, whereas the other two valves lie between ventricles and arteries.

When both atria contract then blood moves to the right and left ventricles through tricuspid and bicuspid (mitral) valves respectively.

INVESTIGATE

At rest our heart beats about 70 to 75 times per minute. What would happen to our heart beats when we exercise?

Teacher Note: The teacher should show the model of human circulatory system. Also ask students question related to main components of system during observation. Teacher should also help students to correctly identify the location of heart and blood vessels, and draw the picture of the system in their notebooks.

When ventricles contract, then blood moves to the right pulmonary artery and left aorta through pulmonary and aortic valves respectively. Pulmonary artery takes the de-oxygenated blood towards lungs for purification, whereas aorta transport oxygenated blood to all parts of body. Figure 2.5 shows major steps involve in the circulation of blood with in the heart.

Circulation of Blood:

Circulation of blood starts from the right side of the heart. Heart pumps blood out of it along an artery to the lungs. Lungs purify the blood by diffusing oxygen from the inhaled air and release carbon dioxide in the exhaled air.

Through pulmonary vein' blood returns to the heart. (Fig:2.4)

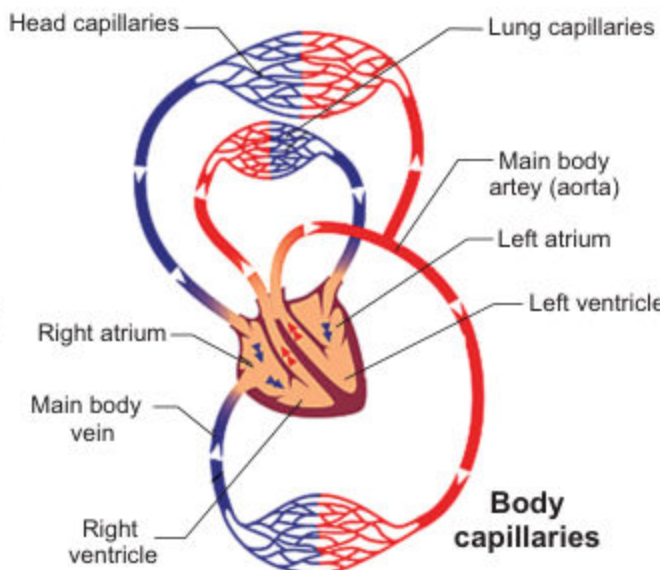
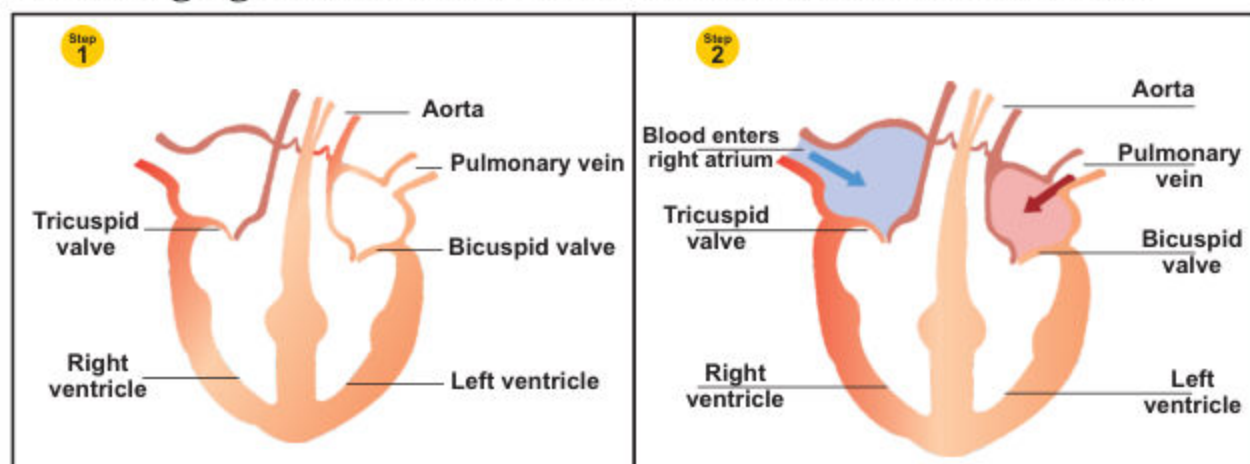


Figure 2.4 Circulation of blood in human

With the help of the pumping action of the heart, blood now enters in an aorta which transports blood to all body parts. Near cells, arteries spilt into smaller blood vessels called capillaries. In capillaries exchange of gases, digested food, minerals and waste products between cells and blood occur. The oxygen from the blood diffuses into the cells and carbon dioxide produced after respiration, diffuses from the cells into the blood. Hence in capillaries oxygenated blood changes into de-oxygenated blood. The capillaries join together to form veins carrying de-oxygenated blood, back to the right side of the heart.

Teacher Note: The teacher should show the video/ chart of human circulatory system. Also ask students questions related to role of blood vessels and heart in blood circulation. Teacher should help students to make a model of human circulatory system and present it in front of the class.

Following figures show circulation of blood within the heart.



Right atrium receives blood from the body parts through Vena cava and left atrium from the lung through pulmonary vein

Four chambers of heart

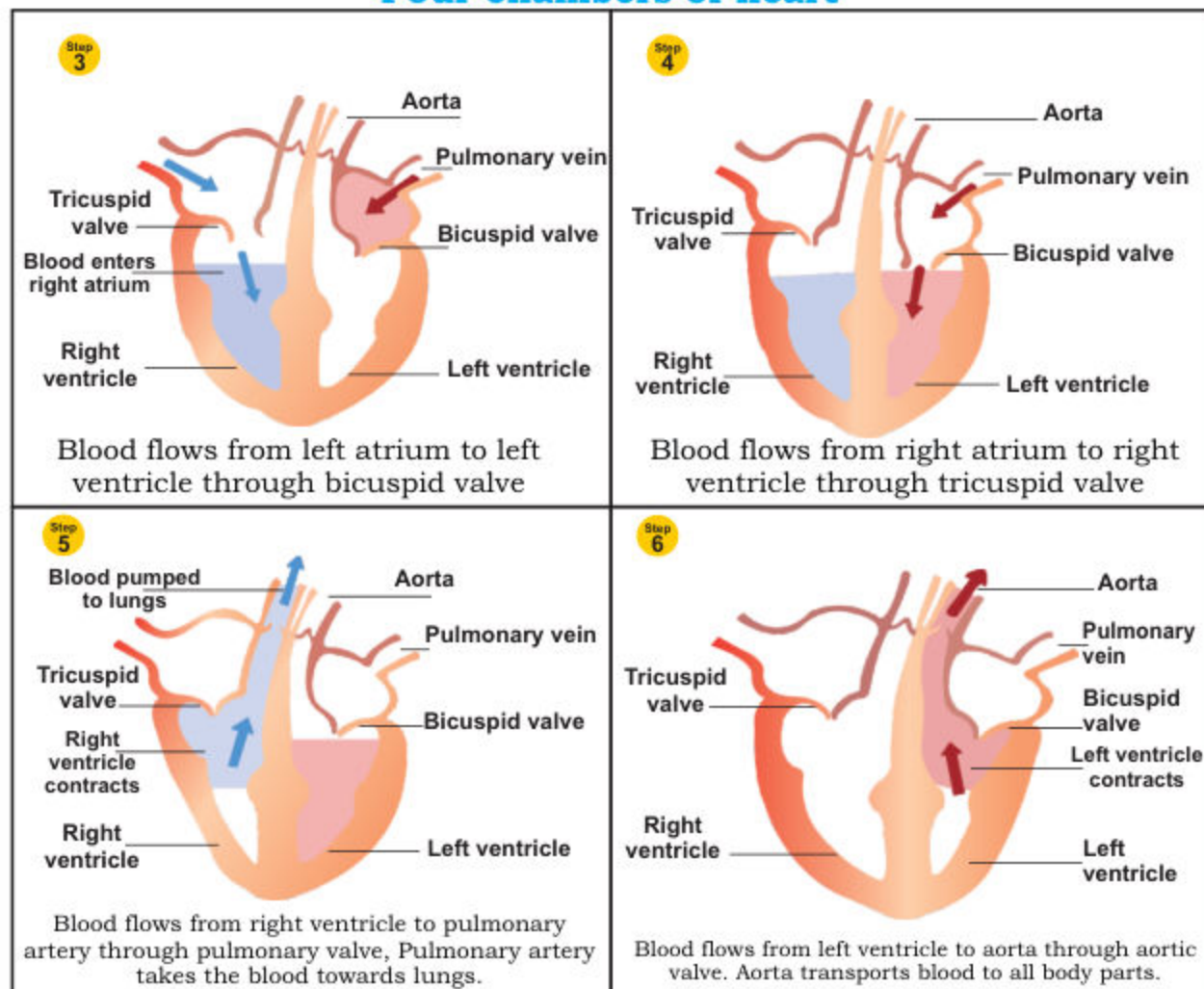


Figure 2.5 Circulation of blood within the heart

Activity 2.1: To investigate the effect of physical exercise on pulse rate.

What I need:

- Stop watch

What to do:

- Hold one of your hands out with the palm facing upwards and the elbow slightly bent.
- Place your index and middle fingers on the inside of your wrist, just below the base of your thumb, as shown in figure 2.6.
- Press the two fingers lightly on your skin until you feel your pulse.
- If you feel nothing, either press harder or search with your fingers for the artery.
- Use stop watch to note down the time.
- Count the number of pulses for one minute.
- Now go outside the classroom and run for 2 to 3 minutes.
- Come back to your classroom and note down your pulse rate as per the previous method.



Figure 2.6 Checking pulse

What I observed:

Activity Questions:

1. Were you able to feel your pulse on wrist?
2. Did you note time properly?
3. What was the pulse count in a minute?
4. What was the pulse count after fast walk or running?
5. Did you feel any difference in these two pulse rates? Why?

What I concluded:

Teacher Note: Teacher should help students to feel their pulse and let them to note down the time for a minute then send them outside the classroom for fast walk or running.

DISORDERS IN HUMAN TRANSPORT SYSTEM AFFECTED BY DIET

- ✓ Find out some disorders in human transport system that can be affected by diet.

Diet quality is strongly related to the diseases of circulatory system (cardiovascular system) in humans. It involves the heart and blood vessels. There are different heart diseases such as angina, heart attack, hypertension and hypotension.

People taking high quantity of fat, salt and low quantity of fruits, vegetables, whole grain and fish can have more risk of cardiovascular diseases. Besides this, the amount of salt consumed is also important to raise the risk of blood pressure and other cardiovascular diseases. Frequent intake of fast foods, processed foods and meat that are high in fats and sugars, promote obesity and may increase cardiovascular risk.

DO YOU KNOW?

- ❖ Hypertension may also cause brain stroke and haemorrhage.
- ❖ To avoid hypertension, it is important to take following preventive measures:
 - Do not become over weight.
 - Do not smoke.
 - Avoid stress and tension.
 - Avoid high cholesterol food.

Hypertension and Hypotension:

Hypertension is also known as high blood pressure. It is a chronic medical condition in which the blood pressure in the arteries is persistently increased. Blood pressure is expressed by two measurements; maximum and minimum pressures. Normal resting blood pressure in an adult is approximately 120/80 mm Hg. If the blood pressure is below the normal, the condition is Hypotension or low blood pressure. Usually people do not consider it as disorder but this condition can also cause havocs. As most of the symptoms are common in both the conditions of blood pressure, so it is advised to check the blood pressure before having any medicine.

Angina:

Angina is chest pain that occurs if an area of heart muscle does not get enough oxygen-rich blood. It is a symptom of coronary heart disease.

Heart Attack:

Heart attack is a condition in which narrowing of arteries occur due to the deposition of high level of fatty acids (cholesterol) in arterial walls which increases the blood pressure. These arteries supply oxygen-rich blood to heart.

Due to accumulation, blood flow through a coronary artery gets mostly blocked. If blood flow isn't restored quickly, the section of heart muscle begins to die.

The symptoms of a heart attack can vary from person to person. Some of the most common symptoms of a heart attack are:

- Chest pain or discomfort.
- Upper body discomfort.
- Shortness of breath.
- Breaking out in a cold sweat.
- Feeling unusually tired for no reason, sometimes for days.
- Nausea and vomiting.
- Light headaches.

Heart attack can be diagnosed by different tests such as electrocardiogram (ECG), blood test and echo-cardiography (generally called as echo test). Heart attack may not occur if you have healthy lifestyle and good relationship among your family members.

Diabetes:

Diabetes is the common disorder of endocrine hormones system. In diabetes, blood glucose (sugar) level is high. Diabetes causes excessive thirst and large amount of urine. In this condition body does not process food properly for use as energy. Diabetes can cause serious health issues, heart disease, kidney failure and blindness.

DO YOU KNOW?

- ❖ Heart attack is scientifically known as myocardial infarction.
- ❖ Despite accumulation of cholesterol, there are many causes of heart attack. Some are as follows:
 - Smoking
 - Increased stress
 - Lack of physical activity
- ❖ The function of heart may be corrected by the use of artificial pacemaker. Pacemaker maintains the normal contraction and relaxation of atria and ventricles.

People with diabetes might have some of following symptoms:

- Frequent urination
- Excessive thirst
- Weight loss
- Extreme hunger
- Sudden vision change
- Feeling tiredness

Blood sugar level is maintained through insulin injections in diabetic patients. Adopting healthy daily practices and taking exercise is very helpful in maintaining sugar level.

TRANSPLANTATION OF DYSFUNCTIONAL BODY PARTS

- ✓ Identify scientific developments that provide alternatives for dysfunctional body parts such as artificial tissues and organs; and their transplantation.

There are developments in the medical field and many approaches have been adopted for treating the patients. One of the developments is formation of artificial tissues and organs and their transplantation.

Do you know what is transplantation?

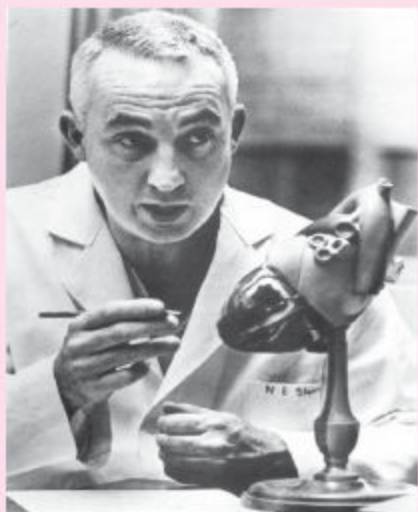
What are the benefits of transplantation to the patient?

“Transplantation is the replacement of a dysfunctional organ with a healthy one.”

Heart transplant replaces the unhealthy patient's heart with a healthy donor heart. As transplantation is a very complicated process, patient requires regular check-ups by specialized cardiologist. However, what would happen if donors are not available? To avoid this problem artificial organs have been developed. An artificial heart maintains the patient's heart blood circulation.

DO YOU KNOW?

- ❖ Norman Shumway from United States is considered as the father of heart transplantation.



- ❖ National Heart Institute (United States) developed the artificial heart program in 1964, successfully implanted the artificial heart in 1969.

TRANSPORT IN PLANTS

- ✓ Describe absorption of water in plants through roots.
- ✓ Explain how the structure of root, stem and leaves of a plant permit the movement of food, water and gases.

Have you ever thought how transportation of materials occur in plants? Do you know some plants are small in size whereas others are very tall, like cone plants? Is the transportation in all plants either small or tall, similar? Do plants also need a pump to transport materials like animals which have heart to perform the job?

The two important processes in plants are photosynthesis and respiration that continuously require food and water.

Do plants efficiently transport food and water as per required?

The transport system in plants consists of xylem and phloem.

The xylem transports water and minerals from roots to the leaves via the stem while phloem transports sugar produced in the leaves to all parts of the plants.

Unscramble the given words
LMYXE

HINT: A tissue transports water in plants
MHPOLE

HINT: A tissue transports food in plants

TRANSPORT EXPLORATION

Explore the role of osmosis and diffusion in plant transportation.

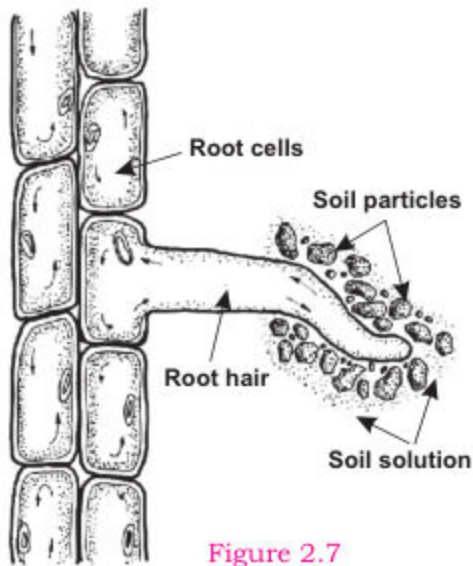


Figure 2.7

Structure of a root hair cell

Absorption of Water in Plants through Roots:

The minerals and dissolved gases available to plants for absorption are dissolved in the soil. The water enters the plant body through its roots. Roots are often extensive and grow rapidly in the soil.

In roots, water absorption takes place through the root hairs. Root hairs are tubular outgrowths of the outermost layer of root. There are thousands of tiny root hairs on each root. Being long, narrow and single cellular structure, the root hair increases the surface area for efficient absorption of water and mineral salts from the soil.

Water enters the root hair by osmosis in which the movement of water molecules from a region of higher concentration (soil) to a region of low concentration (root hairs) through a semi-permeable membrane (cell membrane) occurs.

Movement of water through roots, stem and leaves:

Through the process of osmosis and diffusion, the water and dissolved minerals and gases in the roots move from cell to cell towards the xylem. The xylem then carries them, to all parts of plant through root pressure and transpiration pull. Before discussing these forces let's explore the structure of xylem.

The xylem vessels form continuous channel from the roots through stem to the leaves. Water can move up through these channels. Xylem vessels are composed of dead cells, joined together to form long tubes. The strength in xylem vessel is provided by cellulose (carbohydrate).

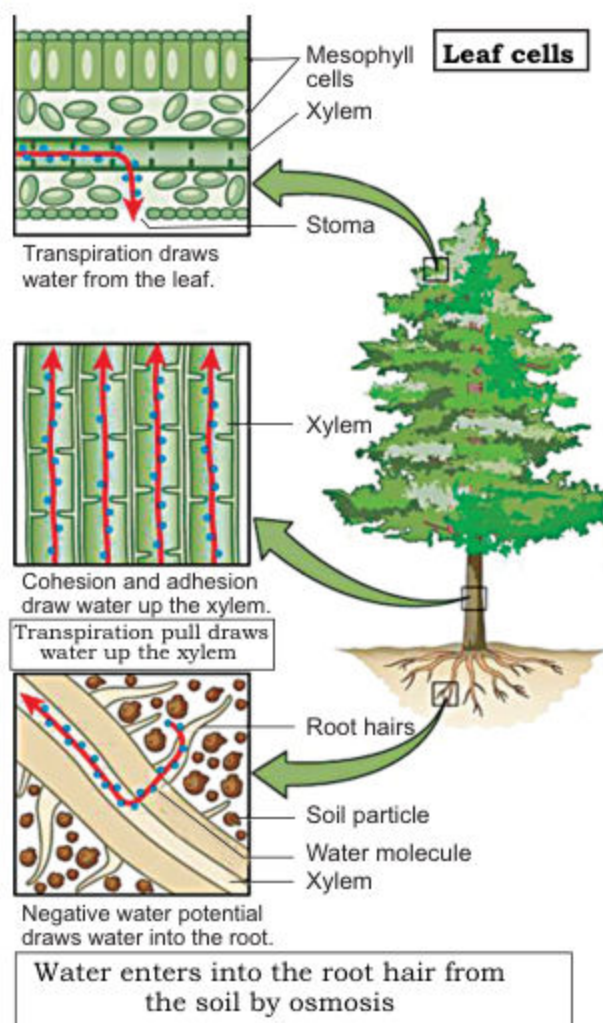


Figure 2.8 Illustrates the process of transpiration pull. (Transpiration of water in plant through xylem)

Root Pressure:

A force that is involved in the movement of water and dissolved minerals up in the xylem is called the root pressure. As a result of root pressure, the water with dissolved minerals rise to desired height in most plants.

Also it is not sufficient to bring water up to the leaves in tall plants.

How is water transported in tall plants?

Transpiration pull:

In tall plants transpiration pull transports the water in xylem. As we discussed in our previous class the process of transpiration.

“Transpiration is the evaporation of water from the aerial parts of the plant especially through leaves (stomata: small opening in leaf).”

Due to transpiration, transpiration pull occurs.

When leaf transpires the water inside the leaf cells drops.

This drop causes water to move from xylem (high water level) to leaf cells (low water level) through osmosis. The water then leaves the xylem thus generating the pulling force all the way to the root where water is pulled from the xylem.

TRANSPORT EXPLORATION

Transportation in xylem is unidirectional whereas in phloem is bidirectional. Why?

Activity 2.2: How is water transported in plants?

What I need:

- Water Glasses (3-4)/Beakers.
- Water
- Food colour
- Leaves of balsam plants
- Sharp knife
- Cutting board

What to do:

- Fill half each of the water glass with water.
- Add few drops of red food colour in each of the water glasses.
- Stir the colour into the water.
- Cut the lower end of the balsam plant.
- Immediately put the roots of balsam plant in the water with colour.
- Leave it in the sun light.
- Observe them after 24 hours, and 48 hours.

What I observed:

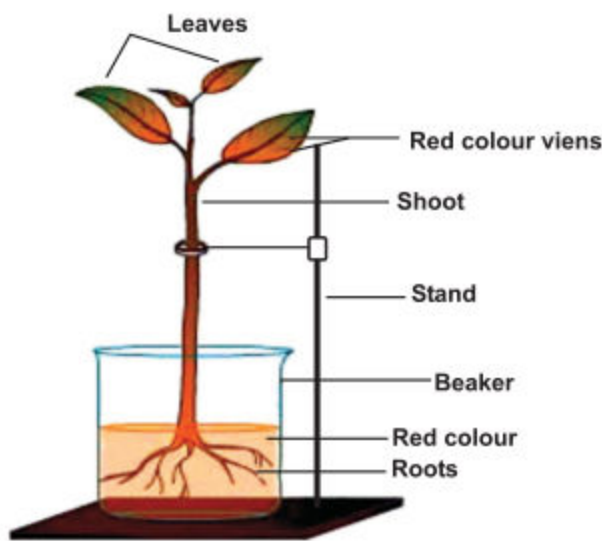


Figure 2.9

Roots are immersed in ink

Activity Questions:

- Why did you cut the stem at the lower end?
- Why did you use food colour?
- Which part of plant stained red?

What I concluded:

TRANSLOCATION IN PLANTS

Translocation is the movement of food materials from leaves to other tissues throughout the plant. We have studied in class six that plants produce food (sucrose: carbohydrates) in their leaves by the process of photosynthesis, but other parts of the plant also require carbohydrates. For this reason, food is translocated from leaves to roots, stems, flowers and fruits. The tissue in which food moves, is the phloem.

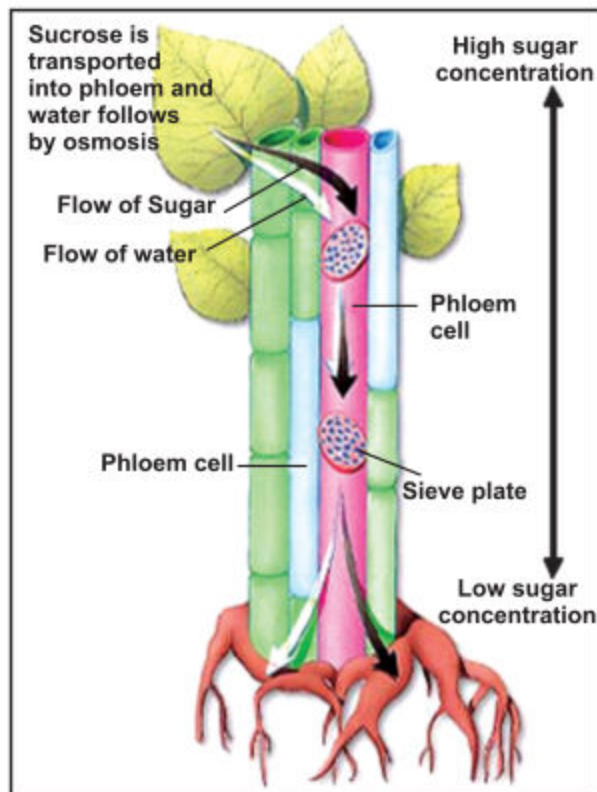
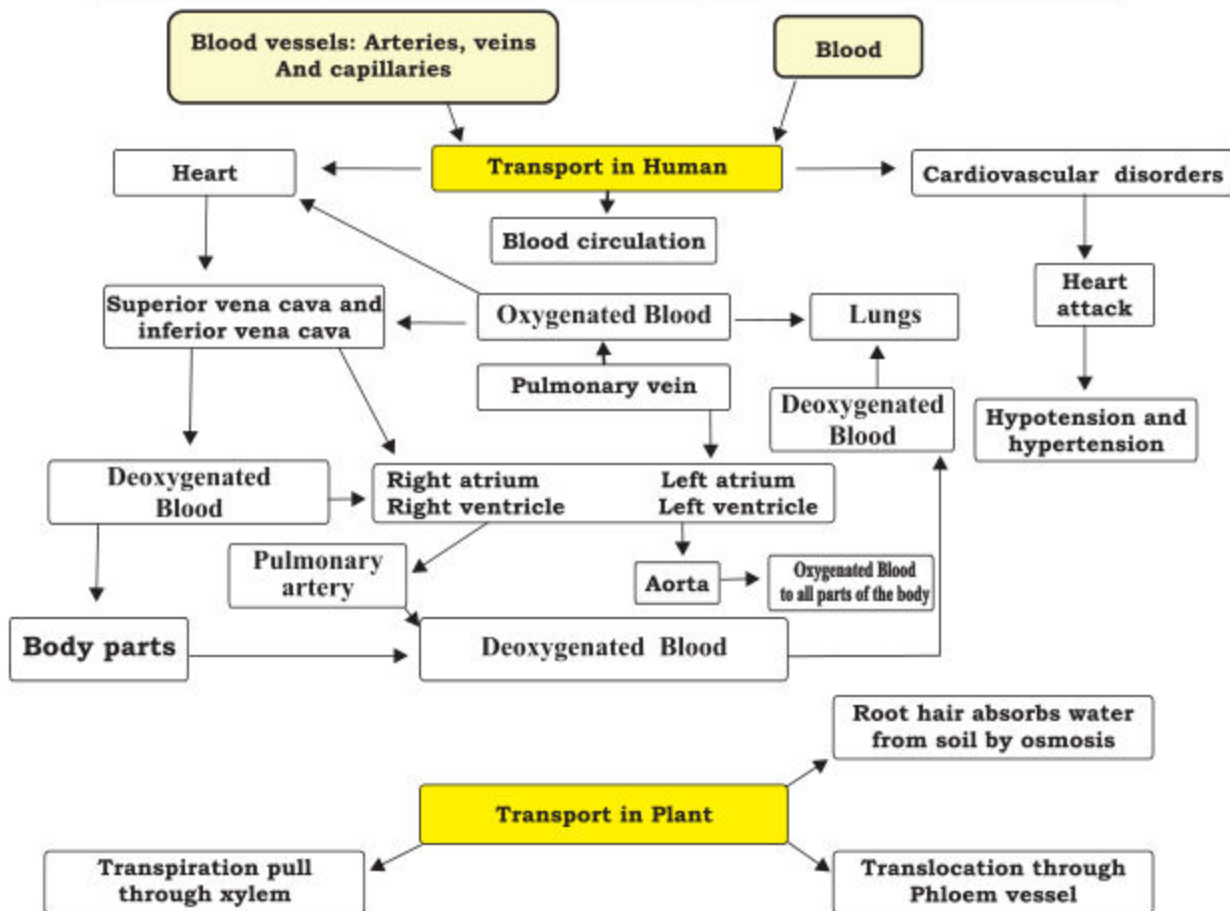


Figure 2.10 Translocation of food in plant through phloem

The phloem is long, continuous channel that extends through the roots and stem and reach into the leaves as veins. The food moves in phloem is composed of sugar dissolved in water, also called sap. All plants translocate carbohydrates. In many others, organic compounds are found such as proteins and hormones.

Summary

	Arteries	Veins
Direction of Blood Flow	From the heart to the body	Back to the heart
Oxygen Concentration	Carry oxygenated blood (except pulmonary artery)	Carry deoxygenated blood (except pulmonary veins)
Pressure	High	Low
Structure	Thick outer walls, elastic and muscle layers	Thin outer walls, elastic and muscle layers
Location	Deeper in the body	Closer to the skin
Valves	No	Yes



Review Exercises

1. Match the statement of column 'A' with column 'B'.

Column A	Column B
i) Blood pressure also cause	(a) Xylem
ii) Absorbs water from the soil	(b) Aorta
iii) Occurs due to blockage in arteries	(c) Root hairs
iv) Supplies blood to all the parts of body	(d) Hypertension
v) Transports water in plants	(e) Heart attack

2. Fill in the blanks:

- De-oxygenated blood from all parts of the body is received by _____ in heart.
- Lungs receive de-oxygenated blood from heart through _____.
- The force applied by leaf cells, to drag water from the roots is known as _____.
- The replacement of a dysfunctional organ with a healthy one is _____ of the organs.
- Transportation of water takes place through continuous tubes in roots, stem and leaves are called as _____.

3. Explain why?

- Valves are present in veins.
- Arteries contain oxygenated blood except pulmonary artery.
- Narrowing of arteries cause heart attack.
- The flow of xylem is unidirectional.
- Ventricles have thick muscular walls than atria.

4. Draw a labelled diagram of heart showing flow of blood in it.

5. Differentiate between the following:

- Arteries and veins
- Atria and ventricles
- Oxygenated and deoxygenated blood

PROJECT

Does phloem transport food (carbohydrates) to the different parts of the plant?

Materials required:

- Two potted plants.
- Sharp knife
- Water

Procedure:

1. Take two potted plants.
2. Mark one plant as A and second as B.
3. Remove a part of the stem tissue in the form of a ring or girdle from plant A.
4. Keep the plant B unchanged.
5. Keep both the plants in proper sunlight and water them.
6. Keep observing both the plants after 5 days, 10 days, 15 days, and 20 days or more period.
7. Note the changes over time.

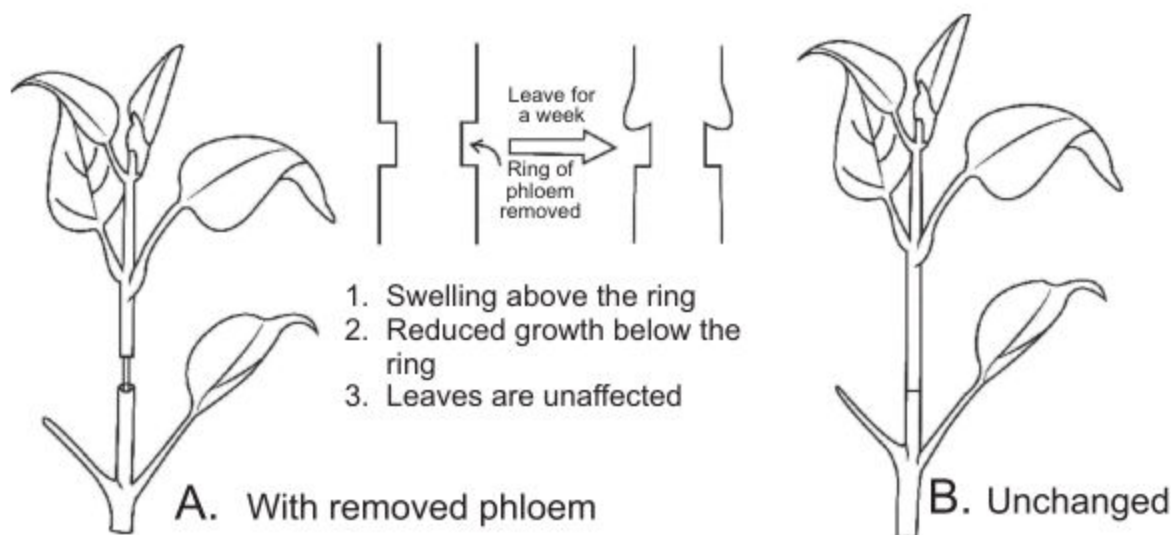


Figure Ringing experiment on phloem detection

Investigating Questions:

- What tissues were being removed along the removal of bark?
- What did you observe on the cut part of the stem?
- What does this experiment tell you about phloem?
- Why plant did not get wilt?
- What was your conclusion?

Explanation:

After few days it will be noticed that in plant A the portion of the stem immediately above the ring bulges (accumulation of food), while no such thing is observed in plant B which was taken as control (for comparison). This is due to the downward translocation taking place through phloem is affected in A due to its (phloem) removal, while no such thing happened in B as the phloem was not removed. This experiment suggests that phloem elements are involved in the translocation of food materials.